RACE
Regular Analysis Through Continuous Exercise

Thermodynamics 1 : RACE-01

Introduction; Thermodynamic terms; Types of the system

1. Thermodynamics mainly deals with

(A) interrelation of various forms of energy and their transformation from one form to another.

(B) energy changes in the processes which depend only on initial and final states of the microscopic systems containing a few molecules.

(C) how and at what rate these energy transformations are carried out.

(D) the system in equilibrium state or moving from one equilibrium state to another equilibrium state.

CTF158

The State of the system; State function; Path function

2. Which of the following is not a thermodynamic property of a system?

(A) H (B) P (C) E (D) w

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3. The volume, V, of a right circular cylinder is given by $V = \pi r^2 h$, where r is the radius and h is the height of cylinder. Show that

(a) $dV = \pi r^2 dh + 2\pi rh\, dr$

(b) V is state function

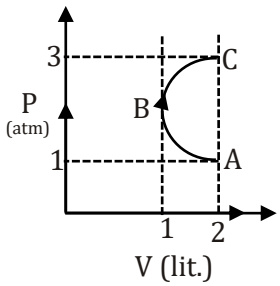
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Work Done

4. In the PV-diagram shown in figure,

Find the work done in the process ABC (in *l*-atm)



CTF162

Heat and Heat Capacity (Cm, Cs, Cp, Cv)

5. In an exothermic reaction, heat is evolved, and system loses heat to the surrounding. For such system

(A) q_p will be negative (B) $\Delta_r H$ will be negative

(C) q_p will be positive (D) $\Delta_r H$ will be positive

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Enthalpy

6. Which of the following statement is/are true for one mole of an ideal gas ?

(A) $\left(\frac{\partial H}{\partial T}\right)_P - \left(\frac{\partial U}{\partial T}\right)_V = R$ (B) $\left(\frac{\partial H}{\partial T}\right)_P > \left(\frac{\partial U}{\partial T}\right)_V$

(C) $\left(\frac{\partial H}{\partial V}\right)_T = 0$ (D) $\left(\frac{\partial H}{\partial P}\right)_T = 0$

CTF161

First Law of Thermodynamics

7. A system absorbs 20 kJ heat and does 10 kJ of work. The internal energy of the system

(A) increases by 10 kJ (B) decreases by 10 kJ

(C) increases by 30 kJ (D) decreases by 30 kJ

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8. The volume of a system becomes twice its original volume on the absorption of 300 cal of heat. The work done on the surrounding was found to be 200 cal. What is ΔU for the system?

(A) 500 cal (B) 300 cal (C) 100 cal (D) −500 cal

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9. A system is provided 50 J of heat and work done on the system is 10 J. The change in internal energy during the process is :-

(A) 40 J (B) 60 J (C) 80 J (D) 50 J

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Degree of freedom

10. The number of degrees of freedom of molecules in a gas whose molar heat capacity is 29 J/mol-K in the process PT = constant is

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ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	AD	D	use Euler theorem	$\left(-\frac{\pi}{2}\right)$	AB	ABCD	A	C
Question	9	10						
Answer	B	3						

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Thermodynamics 1 : RACE-02

Poisson's ratio

1. For which of the following gases, the difference in specific heats at constant pressure and at constant volume is equal to 0.04545 cal/gm?

(A) N_2 (B) N_2O (C) CO (D) CO_2

CTF177

Isothermal Process

2. Which of the following is the best container for gas during isothermal process?

(A) Glass vessel (B) Wood vessel (C) Copper vessel (D) Thermos flask

CTF178

3. Mark as true or false with appropriate reason -

Assertion	T/F	Reason
1. ΔH is a state function		
2. C_V is independent of T for a perfect (ideal) gas.		
3. $\Delta U = q + w$ for every thermodynamic system at rest in the absence of external field.		
4. A process in which final temperature equals initial temperature must be an isothermal process		
5. U remain constant in every isothermal process in a closed system		
6. q = 0 for every cyclic process.		
7. $\Delta U = 0$ for every cyclic process.		
8. A thermodynamic process is specified by specifying only initial and final state of system.		
9. P-V work is usually negligible for solid and liquid.		
10. If neither heat nor matter can enter or leave a system, that system must be isolated.		

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4. A balloon containing air at 27°C and 1 bar initially is filled with air further, isothermally and reversibly till the pressure is 4 bar. If the initial diameter of the balloon is 1 m and the pressure at each stage is proportional to diameter of balloon, the magnitude of work done (in 10^7 J) is

CTF179

5. The final volume (in L) of one mole of an ideal gas initially at 27°C and 8.21 atm pressure, if it absorbs 420 cal of heat during a reversible isothermal expansion is (ln 2 = 0.7)

CTF180

Isochoric process

6. Three moles of an ideal gas being initially at a temperature 273 K was isothermally expanded 5.0 times its initial volume and then isochorically heated so that the pressure in the final state became equal to that in the initial state. The total amount of heat transferred to the gas during the process equals 80 kJ. The value of $C_{v,m}$ for this gas (in cal/K-mol) is (ln 5 = 1.6)

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Isobaric process

7. A quantity of 70 calories of heat is required to raise the temperature of 2 mole of an ideal gas at constant pressure from 40°C to 50°C. The amount of heat required to raise the temperature of the same gas through the same range at constant volume is

(A) 90 calorie (B) 70 calorie (C) 50 calorie (D) 30 calorie

CTF182

8. One mole of an ideal gas undergoes the following cyclic process.

(i) Isochoric heating from (P_1, V_1, T_1) to double temperature.

(ii) Isobaric expansion to double volume.

(iii) Linear expansion (on P–V curve) to (P_1, $8V_1$).

(iv) Isobaric compression to initial state.

If T_1 = 300 K, then the magnitude of net work done by the gas in the cyclic process is

(A) 2400 cal (B) 1200 cal (C) 4800 cal (D) 3600 cal

CTF183

9. A quantity of 8 g oxygen gas is expanded isothermally at 27°C from 2 dm^3 to 8 dm^3 at a constant external pressure of 4 bar. If the magnitude of work done in this process is used in lifting a body of mass 40 kg, the height (in metre) up to which the body can be lifted is (g = 10 ms^{-2})

CTF184

10. Ten moles of a certain ideal gas at a temperature 300 K was cooled isochorically so that the gas pressure reduced 2.0 times. Then, as a result of the isobaric process, the gas expanded till its temperature got back to the initial value. The total amount of heat absorbed (in Kcal) by the gas in this process is

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	BD	C	**1.** False **6.** False **2.** False **7.** True **3.** False **8.** False **4.** False **9.** True **5.** False **10.** False	1	6	5	D	A	6	3

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Thermodynamics 1 : RACE-03

Adiabatic process

1. Which of the following is true for reversible adiabatic process involving an ideal gas?
(A) Gas with higher γ has high magnitude of slope in a P (y-axis) vs. T (x-axis) curve.
(B) Gas with higher γ has high magnitude of slope in a V (y-axis) vs. T (x-axis) curve.
(C) Gas with higher γ has high magnitude of slope in a P (y-axis) vs. V (x-axis) curve.
(D) Gas with higher γ has low magnitude of slope in a P (y-axis) vs. T (x-axis) curve.

CTF186

Comparison between Isothermal and Adiabatic process

2. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The Poisson's ratio of gas is
(A) 3/2 (B) 7/2 (C) 5/3 (D) 9/7

CTF187

3. In the reversible adiabatic expansion of an ideal monoatomic gas, the final volume is 8 times the initial volume. The ratio of final temperature to initial temperature is
(A) 8 : 1 (B) 1 : 4 (C) 1 : 2 (D) 4 : 1

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4. Two moles of an ideal gas (γ= 1.4) was allowed to expand reversibly and adiabatically from 1 L, 527°C to 32 L. The molar enthalpy change of the gas is
(A) −4200R (B) −2100R (C) −1500R (D) −3000R

CTF189

5. Equal moles of He, H_2, CO_2 and SO_3 gases are expanded adiabatically and reversibly from the same initial state to the same final volume. The magnitude of work is maximum for (Assume ideal behaviour of gases and all the degree of freedoms are active.)
(A) He (B) H_2 (C) CO_2 (D) SO_3

CTF190

6. Two moles of an ideal gas (Cv,m = 1.5R) is subjected to the following changes in states.

$$A\ (500\ K, 5\ bar) \xrightarrow[\text{Expansion}]{\text{Reversible isothermal}} B$$

$$\xrightarrow{\text{Isochoric cooling}} C(250K, 1bar) \xrightarrow[\text{Compression}]{\text{Single stage a adibatic}} D\ (3\ bar)$$

The correct statement(s) is/are
(A) The pressure at B is 2.0 bar.
(B) The temperature at D is 450 K.
(C) ΔH_{CD} = 1000R
(D) ΔU_{BC} = 375R

CTF191

Polytropic process

7. An ideal gas undergoes a process in which its pressure and volume are related as PV^n = constant, where n is a constant. The molar heat capacity for the gas in this process will be zero if
(A) $n = \gamma$ (B) $n = \gamma - 1$ (C) $n = \gamma + 1$ (D) $n = 1 - \gamma$

CTF192

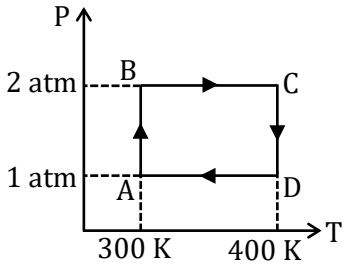
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8. An ideal monoatomic gas undergoes a reversible process, where $\frac{P}{V}$ = constant, from (2 bar, 273 K) to 4 bar. The value of $\frac{\Delta U}{w}$ for this process is

(A) +3.0 (B) −3.0 (C) −1.5 (D) +1.5

CTF193

9. Two moles of helium gas undergoes a cyclic process as shown in the figure. Assuming ideal behaviour of gas, the magnitude of net work done by the gas in this cyclic process is



(A) 0 (B) 100R ln2 (C) 100R ln4 (D) 200R ln4

CTF194

10. An ideal gas is taken reversibly from state A (P, V) to the state B (0.5 P, 2 V) along a straight line in PV diagram. Which of the following statement(s) is/are correct regarding the process?

(A) The work done by gas in the process A to B exceeds the work that would be done by it if the same change in state were performed isothermally.

(B) In the T–V diagram, the path AB becomes a part of parabola.

(C) In the P–T diagram, the path AB becomes a part of hyperbola.

(D) On going from A to B, the temperature of the gas first increases to a maximum value and then decreases.

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	CD	A	B	B	D	ABC	A	B	C	ABD

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